INCH-POUND
A-A-59871
12 July 2010

COMMERCIAL ITEM DESCRIPTION

WATER CLOSET, VACUUM, FOR SHIPBOARD USE

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description (CID) covers vacuum water closets (toilets) used with sewage vacuum, collection, holding, and transfer (VCHT) systems aboard ships.

2. CLASSIFICATION. Vacuum water closets shall conform to the following grades, types, and styles:

2.1 Grades.

Grade A – White vitreous china

Grade B – Stainless steel

2.2 Types.

Type I – Deck-mounted

Type II – Bulkhead (wall) direct-mounted

Type III – Bulkhead (wall) carrier-mounted

2.3 Styles.

Style 1 – Regular bowl

Style 2 – Elongated bowl

3. SALIENT CHARACTERISTICS.

3.1 Description. The vacuum water closet (toilet) shall consist of a toilet bowl, toilet seat, flushing activator, discharge valve, control assembly, water valve, vacuum breaker, flushing water hoses, discharge coupling with clamps, mounting gasket, edge strip (Types II and III only), and carrier (Type III only).

3.1.1 Parts. When specified, individual parts or components shall be furnished in lieu of a complete assembly.

3.2 Materials. The materials for vacuum water closets shall be as specified herein. When a definite material is not specified, the material selected shall be the best commercial marine grade for the purpose intended. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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3.2.1 Corrosion. Vacuum water closet parts in contact with sewage shall be made of materials resistant to corrosion by these wastes. Shipboard sewage contains freshwater or seawater mixed with various concentrations of human wastes, fixture (water closet and urinal) cleaning agents in accordance with S6480-A4-CAT-010 with most of the cleaners belonging to the following chemical component/cleaning families:

a. Quaternary ammonium compounds (CAS Numbers 32426-11-2, 5538-94-3, and 68424-85-1)
b. Inorganic borates
c. Ethoxylated alcohols
d. DPM glycol ether (CAS 034590-94-8)
e. Betaines
f. Alpha olefin sulfonates
g. Alkanolamides
h. Alkanolamines
i. Hydrotropes
j. Hydroxides (Potassium, Sodium, Ammonium)
k. N-Alkyl dimethylamine oxides
l. Acids (sulfamic, muriatic, phosphoric, citric, glycolic)
m. D-Limonene (CAS Number 5989-27-5)
n. Sodium metasilicate pentahydride (CAS Number 6834-92-0)
o. Sodium tripolyphosphate (CAS Number 7758-29-4)
p. Polyglycols
q. Urea (CAS Number 57-13-6)
r. Propylene glycol (CAS Number 57-55-6)
s. Iminoacetic acid (including EDTA)/Chelating Agents

Shipboard sewage in addition to the wastes listed above may contain scale prevention products in accordance with MIL-PRF-32217.

3.2.2 Alternate materials. The manufacturer may use alternate materials for those materials specified herein; however, these materials shall have greater corrosion resistance and improved physical and electrical qualities for the intended use than the materials specified.

3.3 Operating requirements.

3.3.1 Environmental. The vacuum water closet shall operate in the following environments:

3.3.1.1 Fluid. Vacuum water closets shall be able to evacuate fluid containing sewage.

3.3.1.2 Temperature. Vacuum water closets shall operate within temperature variations from 40 to 100 °F, with no effect on operation.

3.3.1.3 Flushing water pressure. Vacuum water closet internal wetted components shall withstand and operate when subjected to continuous internal water pressure equal to maximum water pressure rating of the toilet (not to exceed 100 pounds per square inch) without leakage, failure, or corrosion.

3.3.1.4 Vacuum. Vacuum water closet internal vacuum components shall withstand and operate when subjected to continuous internal vacuum of 8 to 20 inches mercury (Hg) without leakage or failure.

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3.3.1.5 Shock. When specified (see 7.5), the vacuum water closet assembly (stainless steel only) shall be shock tested in accordance with MIL-S-901 and meet the requirements for Grade B, Type A hull/deck-mounted equipment. During shock testing, the vacuum water closet assembly shall be mounted to simulate a shipboard installation.

3.3.1.6 Vibration. The vacuum water closet assembly shall operate within the vibration limits set in MIL-STD-167-1, Type I over the expected service life of the unit (see 3.3.1.8).

3.3.1.7 Airborne noise. When specified (see 7.5), the water closet shall meet noise Requirement 5 of MIL-STD-1474 for Grade B equipment during operation.

3.3.1.8 Service life. The vacuum water closet shall automatically operate for a minimum of 200,000 cycles with no failures or leaks (air or water).

3.4 Design and operation.

3.4.1 Design.

3.4.1.1 Toilet bowl. The toilet bowl for Grade A units shall be a white vitreous china bowl of first quality grade in accordance with ASME A112.19.2M. Suppliers of white vitreous china bowls not made in accordance with ASME A112.19.2M must supply evidence, and the technical authority will make a determination of equivalent quality. The toilet bowl for Grade B units shall be 316L stainless steel. The water closet bowl shall have a regular top contour with a nominal 10-inch wide and 13-inch long opening; a Type III toilet bowl nominal opening shall be 15 inches long. The toilet bowl configuration and other dimensions except as specified herein shall be in accordance with the contractor's normal practice. The toilet bowl houses the discharge valve, control assembly, and water valve. The toilet bowl is provided with a rear removable cover for access to these components.

3.4.1.2 Toilet seat. Water closet seats shall be white plastic in accordance with ANSI Z124.5, hinged at the rear. The type of seat shall be as specified (see 7.5):

a. Open front without cover.

b. Open front with cover.

c. Closed front with cover.

3.4.1.3 Flushing activator. Flushing activators shall be integral to the water closet or remote pushbuttons. A pushbutton on the water closet or remote pushbutton shall be provided that requires manual activation to initiate the flush sequence. The external pushbutton shall be provided with means to attach the device to the bulkhead.

3.4.1.4 Discharge valve. The discharge valve shall be a pneumatically power-operated tubular diaphragm valve. The valve shall automatically close upon loss of system vacuum or failure of the valve bellows or water closet control components. The tubular diaphragm shall be made of neoprene rubber.

3.4.1.5 Control assembly. A control assembly shall be provided that controls operation of vacuum water closet components. Tubing from the control assembly to the vacuum collection system piping shall be provided with a check valve to prevent contamination of the control assembly.

3.4.1.6 Water valve. The flushing water supply valve shall be a pneumatically-operated diaphragm valve. The valve shall dispense the amount of flushing water specified in 3.6.3 when activated.

3.4.1.7 Vacuum breaker. A U.S. Public Health Service (PHS)-approved vacuum breaker or other anti-siphon device shall be provided to prevent contaminated water from being back-siphoned into the flushing water supply. The device may be separately mounted external to the water closet. Separately mounted vacuum breakers shall be provided with means to attach the device and connecting pipe to the bulkhead.

3.4.1.8 Flushing water hoses. Flexible hoses with ½-inch pipe connections shall be furnished to connect toilet water supplies to the vacuum breaker and flushing ring.

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3.4.1.9 Discharge couplings. A 90-degree elbow (neoprene or black PVC) and a straight coupling (neoprene or black PVC) with stainless steel, worm-drive hose clamps shall be furnished. These fittings can be rotated and clamped as required to direct discharge down or to the side.

3.4.1.10 Carrier. A carrier shall be furnished with each Type III toilet to provide the interface between the toilet bowl and both the bulkhead and deck. The carrier shall consist of two brackets capable of being attached to the deck by means of bolts or nuts. The carrier shall be supplied with studs, nuts, and washers as necessary to mount the toilet bowl to the bulkhead.

3.4.2 Operation.

3.4.2.1 Automatic operation. When the flush cycle is initiated by pressing a flush pushbutton, the control assembly (valve or solenoid) shall direct a vacuum-mechanical-hydraulic circuit which opens the flushing water and discharge valves. The flushing water valve shall dispense water to rinse the bowl as atmospheric air propels the bowl contents into the collection piping through the open discharge valve. The discharge valve shall close as it is vented, and additional flushing water shall be directed into the bowl to form a pool of water for the next use.

3.4.2.2 Manual operation. Means shall be provided to manually activate the vacuum water closet flushing cycle for a vacuum water closet with a failed pneumatic pushbutton.

3.4.2.3 Operation after an insufficient vacuum condition. Upon restoration of sufficient vacuum conditions, a water closet shall have the capability to automatically flush if the flush cycle was initiated during the insufficient vacuum condition.

3.5 Dimensions and mountings.

3.5.1 Deck-mounted (type I) toilet. The height of the toilet bowl for deck-mounted toilets, measured at the front top of the rim to the base shall be as shown on figures 1 through 3. Unless otherwise specified (see 7.5), the toilet bowl mounting hole pattern shall be as shown on these figures. The toilet bowl shall be mounted on 5/16-inch studs welded to the deck. Figures 1 and 2 show two alternate outline dimensions for vitreous china toilets, while figure 3 shows dimensions for a stainless unit.

3.5.2 Bulkhead direct-mounted (type II) toilet. The toilet bowl shall be mounted on the bulkhead at the height shown on figure 4. Unless otherwise specified (see 7.5), the toilet bowl mounting hole pattern shall be as shown on this figure. The toilet bowl shall be mounted on 3/8-inch studs welded to the bulkhead or on ship installed supports with 3/8-inch bolts.

3.5.3 Bulkhead carrier-mounted (type III) toilet. The toilet bowl shall be mounted on the bulkhead at the height shown on figure 5. Unless otherwise specified (see 7.5), the toilet bowl mounting hole pattern shall be as shown on this figure. The toilet bowl shall be attached to the bulkhead by 3/8-inch studs that pass through the bulkhead into the toilet carrier brackets on the opposite side of the bulkhead. The toilet bowl shall be secured to the studs with nuts. Bulkhead mounting hardware including studs, flat washers, and nuts shall be furnished with the carrier. The carrier shall be mounted on ½-inch studs welded to the deck.

3.5.4 Discharge connection. The vacuum water closet discharge connection shall be 1½-inch National Pipe Size (NPS). Unless otherwise specified (see 7.5), the discharge connection height above the water closet base or deck level shall be as shown on figures 1 through 5 for the different Grades, Types and Styles of water closets.

3.5.5 Mounting gasket. Each deck-mounted water closet (Type I) shall be furnished with a neoprene gasket having a minimum thickness of 1/8 inch to be mounted on the deck to account for surface irregularities in both the water closet and deck. The gasket material shall be larger to allow the gasket to be trimmed after installation.

3.5.6 Edge strip. Bulkhead-mounted water closets shall be furnished with a white, synthetic rubber, molded edge strip around the periphery of the toilet bowl at the junction with the bulkhead. The edge strip shall be a minimum 1/8-inch thick between the water closet and the bulkhead. The edge strip shall be easy to clean and give a nice appearance.

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3.6 Performance.

3.6.1 Vacuum. The water closet shall operate connected to piping under vacuum ranging from 11 to 18 inches of Hg.

3.6.2 Minimum operating vacuum. The vacuum water closet shall operate at a minimum vacuum of 9 inches Hg.

3.6.3 Flushing water. The water closet shall operate when connected to a freshwater flushing water supply (minimum 30 pounds per square inch, maximum 100 pounds per square inch). The water closet shall use an average of 0.375 gallon of water per flush with an additional 0.125-gallon residual amount of water directed into the bowl.

3.7 Workmanship. Parts shall be of first class workmanship and shall be free from defects of manufacture which detract from their appearance or impair their serviceability.

3.8 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request may be made to the Contracting Officer to determine if the product is acceptable. The Contracting Officer has the option of accepting or rejecting the product.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The item offered must have been sold to the government or commercial market. A modified commercial, off-the-shelf item may be used to minimize design and engineering risk. The offeror shall certify and provide supporting documentation as to the product's serviceability and the long-term (5 years) availability of all necessary maintenance and logistics support parts and materials.

5.3 Bid samples. When specified (see 7.5), one water closet shall be submitted to the Government for examination and testing. The bid sample shall be non-returnable. Shock (see 5.4.9), vibration (see 5.4.11), and airborne noise (see 5.4.10) testing will be performed on the bid sample by the government, as applicable. Failure of the bid sample to meet the requirements of any of the applicable tests or examinations listed herein shall be cause for rejection of the bid.

5.4 Inspection requirements.

5.4.1 Responsibility of inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections, examinations, or tests set forth in this Commercial Item Description where such inspections, examinations, and tests are deemed necessary to ensure supplies and services conform to prescribed requirements.

5.4.2 Failure. Failure of any of the tests and examinations listed herein shall be cause for rejection.

5.4.3 Examination. The vacuum water closet shall be visually examined to determine compliance with all requirements of this CID and with all specified dimensions.

5.4.4 Operational test. The vacuum water closet shall be connected to vacuum and flushing water at specified pressures and operated not less than 10 times to perform automatic flushing operation to demonstrate operation of all moving parts and mechanisms without any leaks.

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5.4.4.1 Flushing water volume. The volume of flushing water discharged during each flush cycle shall be measured to determine compliance with the requirements of 3.6.3.

5.4.5 Minimum vacuum. The vacuum water closet shall be connected to vacuum at the minimum vacuum specified in 3.6.2 and flushing water pressures specified in 3.3.1.3, and operated not less than 10 times to perform automatic flushing operation with no failure to dispense water and discharge.

5.4.6 Manual operation. The vacuum water closet shall be manually operated to demonstrate bypassing all controls to open the discharge valve and evacuate the bowl.

5.4.7 Operation with loss of vacuum. The vacuum water closet shall be activated with zero vacuum in the connected soil line. The water closet shall automatically flush upon restoration of vacuum pressure. This operation shall be demonstrated a minimum of 10 times.

5.4.8 Endurance cycling. The vacuum water closet assembly shall be connected to vacuum and flushing water at system operating pressures and continuously cycled by automatic flush operation for the number of cycles specified in 3.3.1.8 with no failures or leakage. Minor leaks that can be corrected by tightening a connection may be repaired during the test. A failure of controls to operate, mechanism failure, or leak shall be considered a failure and cause for rejection. Certification that this testing requirement has been met may be used in lieu of a test.

5.4.9 Shock test. When shock testing is required (see 7.5), it shall be accomplished in accordance with 3.3.1.5.

5.4.10 Airborne noise test. The noise test, when specified (see 7.5), shall be performed with the water closet connected to vacuum and flushing water at the pressures specified herein, and operated in the automatic discharge mode to determine conformance with requirements specified in 3.3.1.7.

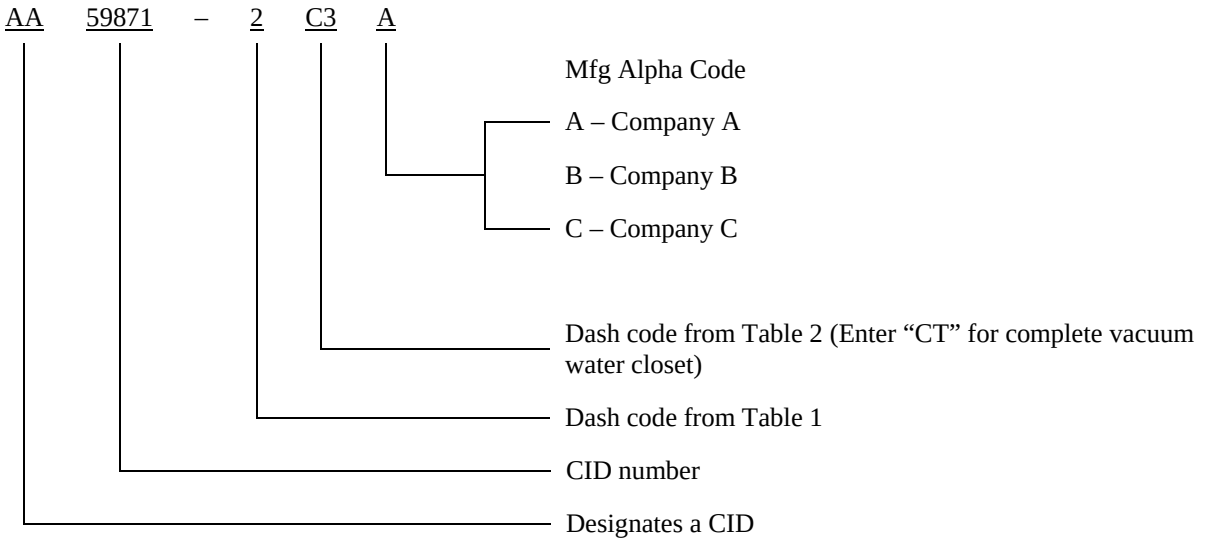
5.4.11 Vibration test. When vibration testing is required (see 7.5), it shall be accomplished in accordance with 3.3.1.6.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see 7.5).

7. NOTES.

7.1 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

Part identification designation



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TABLE I. Dash codes for vacuum water closet.

Dash code	Grade	Type mounted	Style	Seat type	Shock test required	Figure
1	A	I	1	a	No	1
2				b		
3				c		
4	A	I	2	a	No	1
5				b		
6				c		
7	A	I	1	a	No	2
8				b		
9				c		
10	A	I	2	a	No	2
11				b		
12				c		
13	B	I	1	a	Yes	3
14				b		
15				c		
16	B	I	2	a	Yes	3
17				b		
18				c		
19	A	II	1	a	No	4
20				b		
21				c		
22	A	II	2	a	No	4
23				b		
24				c		
25	B	II	1	a	Yes	4
26				b		
27				c		
28	B	II	2	a	Yes	4
29				b		
30				c		
31	A	III	1	a	No	5
32				b		
33				c		
34	A	III	2	a	No	5
35				b		
36				c		

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TABLE I. Dash codes for vacuum water closet – Continued.

Dash code	Grade	Type mounted	Style	Seat type	Shock test required	Figure
37	B	III	1	a	Yes	5
38				b		
39				c		
40	B	III	2	a	Yes	5
41				b		
42				c		

TABLE II. Dash codes for individual components.

Dash code	Individual component	Requirement paragraph	Mfg Alpha Code required
C1	Toilet bowl	3.4.1.1	Yes
C2	Toilet seat a	3.4.1.2.a	Yes
C3	Toilet seat b	3.4.1.2.b	Yes
C4	Toilet seat c	3.4.1.2.c	Yes
C5	Flushing activator	3.4.1.3	Yes
C6	Discharge valve	3.4.1.4	Yes
C7	Control assembly	3.4.1.5	Yes
C8	Water valve	3.4.1.6	Yes
C9	Vacuum breaker	3.4.1.7	Yes

7.2 Source of documents.

7.2.1 American National Standards Institute (ANSI). ANSI standards are available from the American National Standards Institute, 25 W. 43rd St, 4th Floor, New York, NY 10036 or online at http://webstore.ansi.org/.

7.2.2 ASME International. ASME standards are available from ASME International, 22 Law Drive, P.O. Box 2900, Fairfield, NJ 07007-2900 or online at www.asme.org.

7.2.3 FAR. The Federal Acquisition Regulation may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at https://www.acquisition.gov/far.

7.2.4 Federal and Department of Defense specifications and standards. Federal and Department of Defense specifications and standards are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

7.2.5 Naval Sea Systems Command (NAVSEA) Publications. NAVSEA Publications are available from the Naval Logistics Library, 5450 Carlisle Pike, Mechanicsburg, PA 17055 or online at https://nll1.ahf.nmci.navy.mil.

7.3 Sources of supply. Manufacturers whose products are known to meet the requirements of this CID are listed below. Competition is not limited to those products and manufacturers.

EVAC North America, Inc.
1260 Turret Drive
Machesney Park, IL 61115-1486

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7.4 Intended use. Vacuum water closets covered by this CID are intended for use in VCHT systems for new construction ships or as stock replacements for water closets currently installed on existing ships. When acquiring water closets for replacement purposes, the contracting activity must ensure that interface requirements are specified for water, discharge, and electrical connections which are compatible with the water closet being replaced to avoid the need for costly ship modifications. Acquisition for new construction allows for more latitude in water closet dimensions and interface requirements, enabling more suppliers to bid thereby increasing competition. Originally installed water closets may no longer be in production and specifying special mounting dimensions and interface connections for exact replacement should be avoided. Consideration should be given to making minor shipboard changes instead of specifying special dimensions. The availability of supplier-furnished retrofit kits should be investigated. Water, discharge, and electrical connections location and mounting bolt patterns generally are not an issue for new construction.

7.5 Ordering data. The contract or order should specify the following:

a. Title, number, and date of this commercial item description.
b. Grade, type and style of vacuum water closet (see 2).
c. H.I. Shock grade required (see 3.3.1.5).
d. If airborne noise test is required (see 3.3.1.7).
e. Type of toilet seat (see 3.4.1.2).
f. Discharge connection height required (see 3.5.4).
g. If retrofit kit may be provided.
h. Bid samples are required (see 5.3).
i. Levels of preservation, packing, labeling, and marking required (see 6).
j. Mounting hole pattern and size of studs or bolts required (see figures 1, 2, 3, 4, and 5).

7.6 National Stock Numbers (NSNs). The following is a list of NSNs assigned which correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

NSN	ITEM
4510-01-468-4080	5999015-002 (Deck mount, china bowl)
4510-01-112-5789	5999143 (Deck mount, china bowl)
4510-01-299-5799	5700034-002 (Bulkhead mount, china bowl)
4510-01-435-0501	6542578 (Bulkhead mount, china bowl)
4510-01-262-3289	5347001 (Deck mount, stainless bowl)
4510-01-522-8216	7250070 (Deck and bulkhead mount, china elongated bowl)

7.7 Key words.

Fixtures, plumbing
System, pollution control
System, sewage vacuum collection, holding and transfer (VCHT)
Toilet, vacuum

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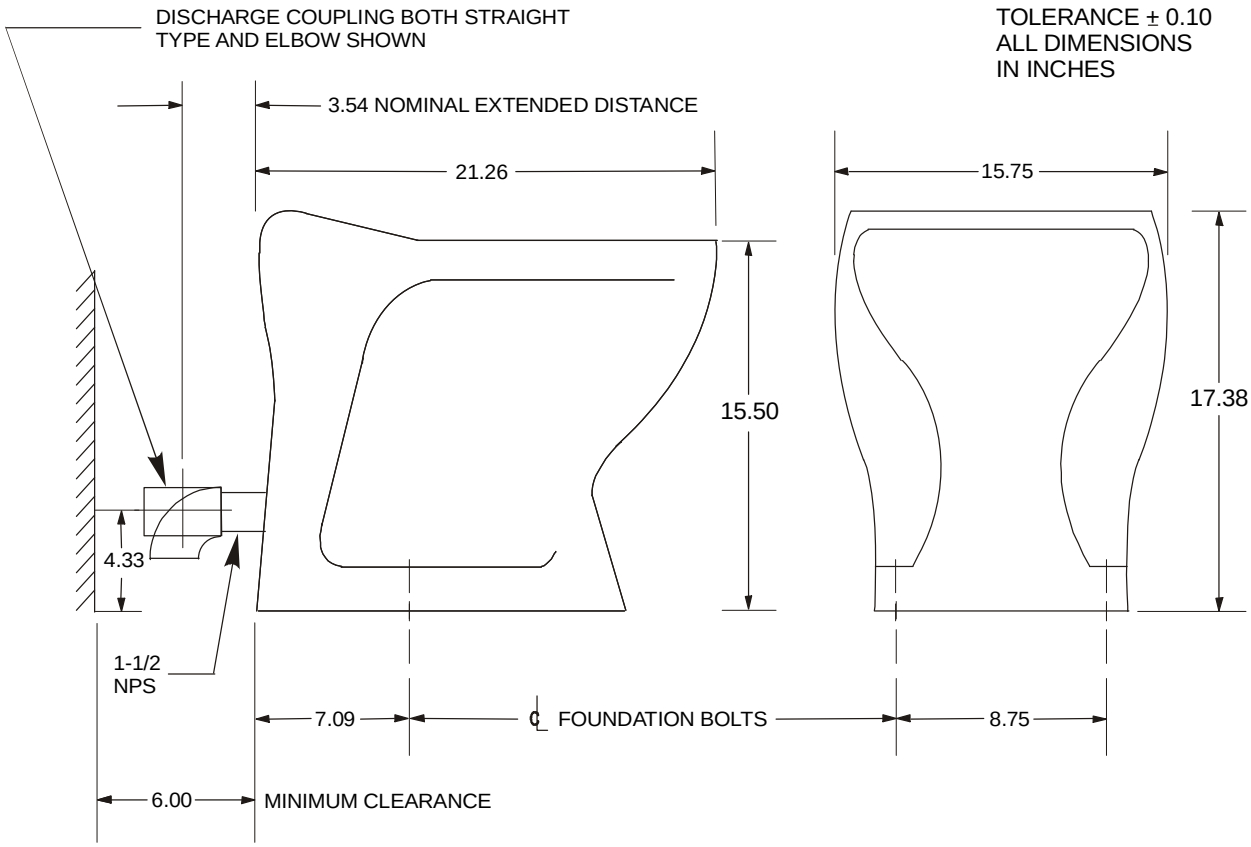


FIGURE 1. Type I, grade A deck-mounted vacuum toilet outline (alternative 1).

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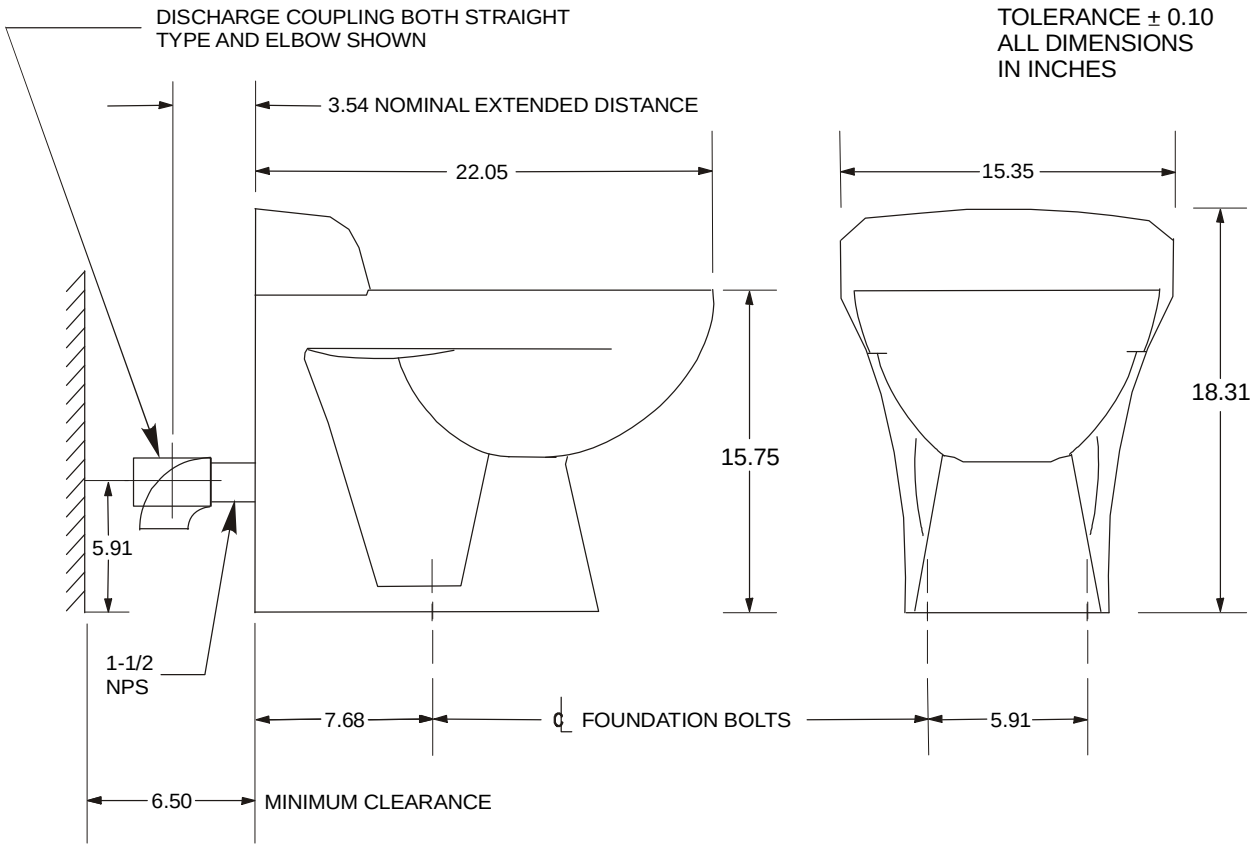


FIGURE 2. Type I, grade A deck-mounted vacuum toilet outline (alternative 2).

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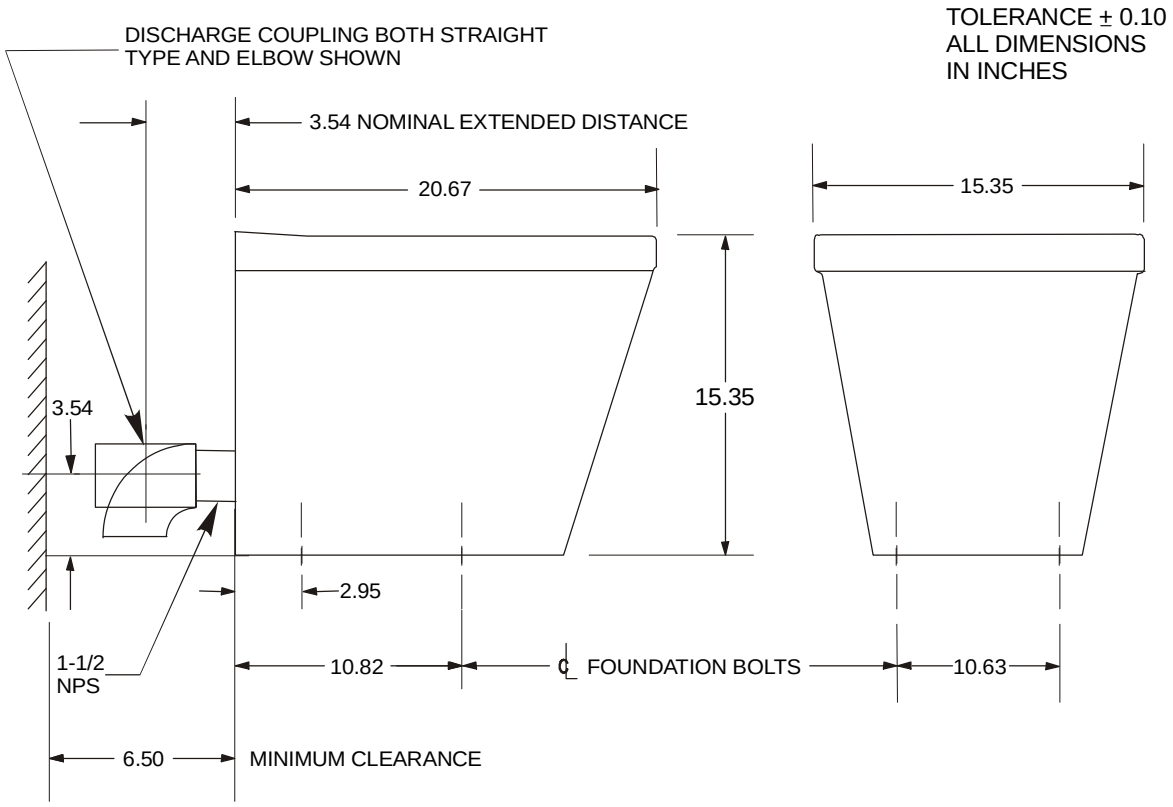


FIGURE 3. Type I, grade B deck-mounted vacuum toilet outline.

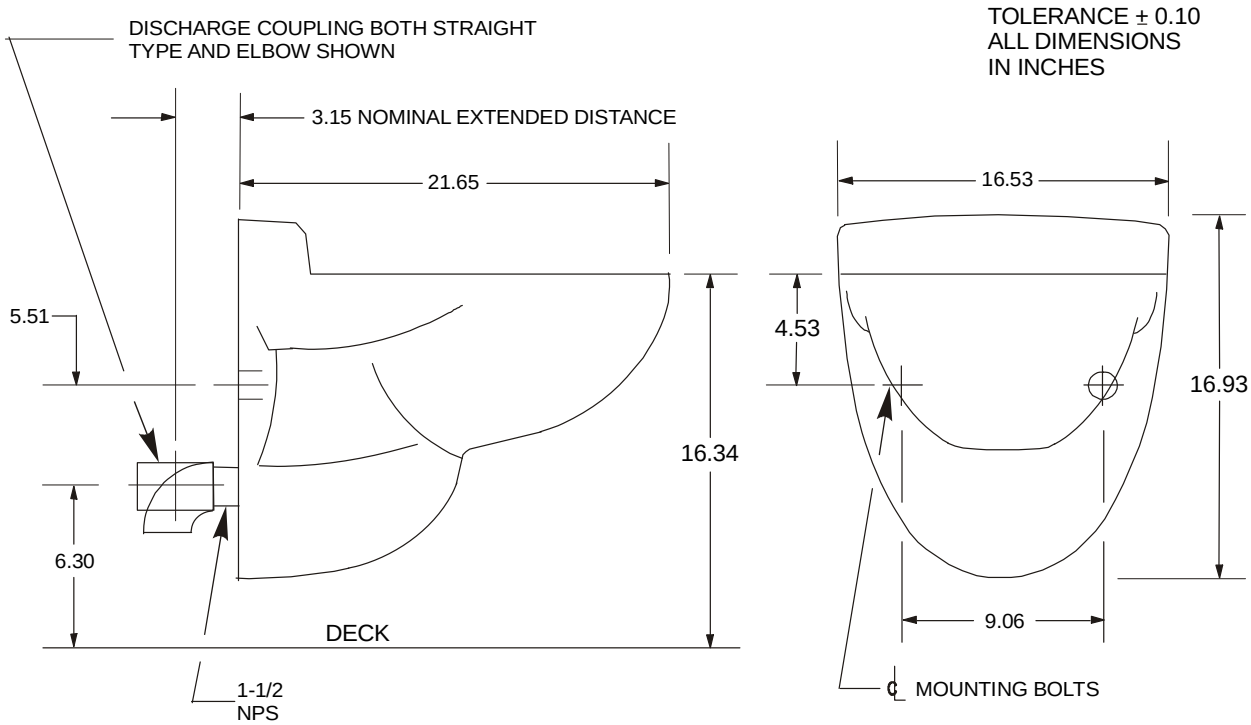


FIGURE 4. Type II bulkhead-mounted vacuum toilet outline.

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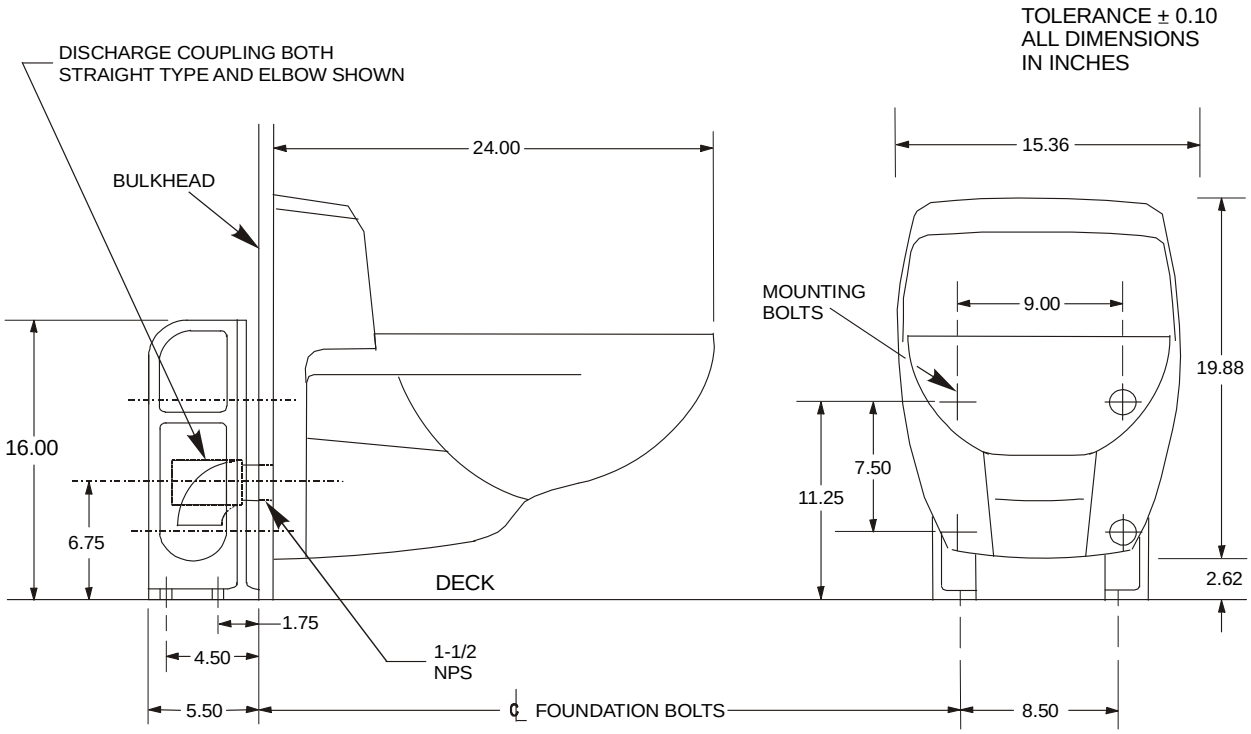


FIGURE 5. Type III, Bulkhead (carrier)-mounted vacuum toilet outline.

MILITARY INTERESTS

Custodian:
Navy – SH

Review Activities:
Navy – YD
DLA – IS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
Navy – SH
(Project 4510-2009-001)

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